

TrES-1b

R-1023

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_161109 · created 2026-07-30T05:02:40+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457701.6085 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.043
Transit depth (Rp/Rs)^2	2.89 +/- 1.45 [%]
Orbital Inclination (inc)	89.84 +/- 1.17
Airmass coefficient 1 (a1)	1.407 +/- 0.032
Airmass coefficient 2 (a2)	-0.1256 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	1.92 %
Best Comparison Star	None
Optimal Aperture	3.82
Optimal Annulus	10.56
Transit Duration (day)	0.0964 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.043 vs 0.1358 (deviation 0.69σ)
Duration fit vs published	0.0964 d vs 0.1042 d (deviation 2.25σ)
Mid-time O–C	-4.1 min
Depth SNR	3.4
Residual scatter	1.92 %

Light curve

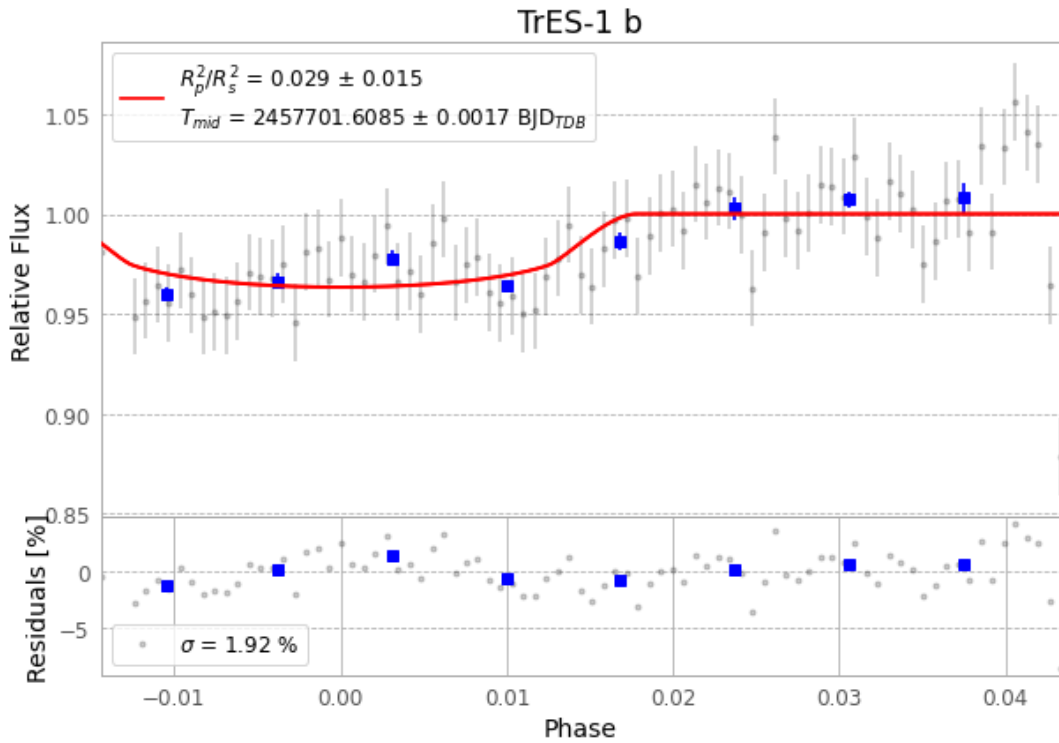


Figure 1 — FinalLightCurve_TrES-1 b_2016-11-08.png (SHA-256 c103bb5a5291ff4e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8464f7eb5a03496a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f8439d0

Data provenance

87 FITS frames (57 MB), TRES-1161109013315.FITS → TRES-1161109055712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-161109.log (2e2d2bdb8319...), FinalLightCurve_TrES-1 b_2016-11-08.png (c103bb5a5291...), FinalLightCurve_TrES-1 b_2016-11-08.csv (256d3b30ef8d...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 05:02Z.